

From free field theory to symmetric polynomials

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Outline

- 1 Representation theory from singular vectors
- 2 Free field theories and screening operators
- 3 Symmetric polynomials
- 4 Application to the minimal models

Def: The universal Virasoro vertex operator algebra $V(c)$, $c \in \mathbb{C}$

- Virasoro algebra: $[L_m, L_n] = (m - n)L_{m+n} + \frac{c}{12}m(m^2 - 1)\delta_{m+n,0}$
- Vacuum rep: $V(c) = U(\bigoplus_{n \geq 2} \mathbb{C}L_{-n})|0\rangle$; $L_m|0\rangle = 0$, $m \geq -1$
- Operator state correspondence:

$$L_{-n-2}|0\rangle \leftrightarrow \frac{\partial^n}{n!}T(z), \quad n \geq 0$$

$$L_{-n_1-2} \cdots L_{-n_i-2}|0\rangle \leftrightarrow: \left(\frac{\partial^{n_1}}{n_1!}T(z) \right) \cdots \left(\frac{\partial^{n_i}}{n_i!}T(z) \right) :$$

- Irreducible if $V(c)$ contains no singular vectors χ :
 $L_n\chi = 0, n \geq 1$; $L_0\chi = N\chi, N \geq 1$.

Representations of this vertex operator algebra include Virasoro Verma modules.

Def: Virasoro Verma module $M(h)$, $h \in \mathbb{C}$

- Generated by ω_h : $L_0\omega_h = h\omega_h$, $L_n\omega_h = 0, n \geq 1$
- $M(h) = U(\bigoplus_{n \geq 1} \mathbb{C}L_{-n})\omega_h$

Minimal models

Let $(p_+, p_-) = 1, p_+, p_- \geq 2$.

Def: The minimal model vertex operator algebra

- $V(c)$ reducible at central charge $c_{p_+, p_-} = 1 - 6 \frac{(p_+ - p_-)^2}{p_+ p_-}$.
- $V(c_{p_+, p_-})$ contains unique ideal $I(p_+, p_-)$ generated by a Virasoro singular vector of conformal weight $(p_+ - 1)(p_- - 1)$.
- The minimal model vertex operator algebra is the irreducible quotient $M_{p_+, p_-} = V(c_{p_+, p_-})/I(p_+, p_-)$.

Cor: Minimal model representation theory

Representations of M_{p_+, p_-} are those of $V(c_{p_+, p_-})$ annihilated by $I(p_+, p_-)$.

Example: Yang-Lee minimal model, $p_+ = 2, p_- = 5$

- central charge $c_{2,5} = -\frac{22}{5}$
- ideal generated by a singular vector χ . $L_0\chi = 4\chi$, $L_n\chi = 0, n \geq 1$

$$\chi = (L_{-2}^2 - \frac{3}{5}L_{-4})|0\rangle$$

$$\chi(z) = :T(z)^2: - \frac{3}{10}\partial^2 T(z) = \sum_{n \in \mathbb{Z}} \chi_n z^{-n-4}$$

$$\chi_0 \omega_h = \oint z^3 \chi(z) \omega_h dz = L_0(L_0 + \frac{1}{5})\omega_h = h(h + \frac{1}{5})\omega_h$$

- Conformal weight of allowed highest weight states can only be 0 or $\frac{1}{5}$.

- Singular vectors **very** hard to find in general.
 - ▶ No general closed formulae known.
- Goal of this talk: Find better way to get highest weights of minimal models by symmetric polynomial magic.
- Later (=David): Why this is not magic and generally applicable.

The free boson

Def: Heisenberg algebra

$$\mathfrak{h} = \bigoplus_{n \in \mathbb{Z}} \mathbb{C} a_n, [a_m, a_n] = m \delta_{m+n, 0}$$

Def: Heisenberg vertex operator algebra (a.k.a. free boson)

- $\mathcal{H} = \mathbb{C}[a_{-1}, a_{-2}, \dots] |0\rangle, a_n |0\rangle = 0, n \geq 0$
- $T = \frac{1}{2}(a_{-1}^2 + \alpha a_{-2}) |0\rangle, c = 1 - 3\alpha^2, \alpha \in \mathbb{C}$
- $a_{-1} |0\rangle \leftrightarrow a(z) = \sum_n a_n z^{-n-1}, T(z) = \frac{1}{2} : a(z)^2 : + \frac{\alpha}{2} \partial a(z)$
- For $(p_+, p_-) = 1, p_+, p_- \geq 2$, let $\alpha_+ = \sqrt{\frac{2p_-}{p_+}}, \alpha_- = -\sqrt{\frac{2p_+}{p_-}}$ and $\alpha = \alpha_+ + \alpha_-$, then $c = 1 - 6 \frac{(p_+ - p_-)^2}{p_+ p_-}$ is central charge of minimal models.
- Minimal models realisable as quotient of Virasoro vertex operator algebra or as **subquotient of Heisenberg vertex operator algebra**.

The free boson

Fock spaces are irreducible $\mathcal{H}$ (and $\mathfrak{h}$) representations.

Def: Fock spaces $F_\lambda, \lambda \in \mathbb{C}$

- $F_\lambda = \mathbb{C}[a_{-1}, a_{-2}, \dots]|\lambda\rangle, a_0|\lambda\rangle = \lambda|\lambda\rangle, a_n|\lambda\rangle = 0, n \geq 1$
- $L_0|\lambda\rangle = \frac{1}{2}(a_0^2 - \alpha a_0)|\lambda\rangle = h_\lambda|\lambda\rangle$
 $h_\lambda = \frac{\lambda}{2}(\lambda - \alpha)$

Def: Vertex operators $|\lambda\rangle \leftrightarrow V_\lambda(z)$

- new operator $\hat{a}$: $[a_n, \hat{a}] = \delta_{n,0}$
- $e^{\lambda \hat{a}}|\mu\rangle = |\lambda + \mu\rangle$
- $V_\lambda(z) = e^{\lambda \hat{a}} z^{\lambda a_0} \prod_{m \geq 1} e^{\lambda \frac{a_{-m}}{m} z^m} \prod_{m \geq 1} e^{-\lambda \frac{a_m}{m} z^{-m}}$
- $T(w)V_\lambda(z) = \frac{h_\lambda}{(w-z)^2} V_\lambda(z) + \frac{1}{w-z} \partial V_\lambda(z) + \text{reg}$
- $V_\lambda(z) : F_\mu \rightarrow F_{\lambda+\mu}((z))z^{\lambda\mu}$

Def: Screening operator [Dotsenko, Fateev]

- If $h_\lambda = 1$ (here $\lambda = \alpha_\pm$), then $S_\pm(z) = V_{\alpha_\pm}(z)$ is called a screening field.
- $T(w)S_\pm(z) = \frac{1}{(w-z)^2}S_\pm(z) + \frac{1}{w-z}\partial_z S_\pm(z) + \text{reg} = \partial_z \frac{S_\pm(z)}{w-z} + \text{reg}$
- Zero mode $S_\pm = \oint_0 S_\pm(z)dz$, called a screening operator, is a Virasoro homomorphism: $[T(w), S_\pm] = \oint_w T(w)S_\pm(z)dz = 0$
- **Caution:** $S_\pm : F_\mu \rightarrow F_{\mu+\alpha_\pm}$ only well defined if $\alpha_\pm \mu \in \mathbb{Z}$.
- If $\alpha_\pm \mu \in \mathbb{Z}$ and $S_\pm|\mu\rangle \neq 0$ then $S_\pm|\mu\rangle$ is a **singular vector** in $F_{\mu+\alpha_\pm}$

Composing screening fields

If $\alpha_{\pm}\mu$ not integer, consider higher powers of the screening operator.

α_+ case: $S_+(z_1) \cdots S_+(z_r) =$

$$e^{r\alpha_+ \hat{a}} \prod_{i \neq j} (z_i - z_j)^{\alpha_+^2/2} \prod_{i=1}^r z_i^{\alpha_+ a_0} \prod_{m \geq 1} \exp \left(\alpha_+ \frac{a_{-m}}{m} \sum_i z_i^m \right) \prod_{m \geq 1} \exp \left(-\alpha_+ \frac{a_m}{m} \sum_i z_i^{-m} \right)$$

Thm: [Tsuchiya, Kanie], [Varchenko]

If and only if $\mu = \frac{1-r}{2}\alpha_+ + \frac{1-s}{2}\alpha_- = \alpha_{r,s}$, $r \in \mathbb{Z}_+$, $s \in \mathbb{Z}$, then there exists a unique choice of a cycle Γ , such that

$$S_+^{[r]} = \int_{\Gamma} S_+(z_1) \cdots S_+(z_r) dz_1 \cdots dz_r : F_{\alpha_{r,s}} \rightarrow F_{\alpha_{-r,s}}$$

is a non-trivial Virasoro homomorphism.

- if $s \geq 1$, then $S_+^{[r]}|\alpha_{r,s}\rangle = 0$.
- if $s \leq 0$, then $S_+^{[r]}|\alpha_{r,s}\rangle \neq 0$.

Composing screening fields: evaluation on $F_{\alpha_{r,s}}$

$$S_+(z_1) \cdots S_+(z_r) \Big|_{F_{\alpha_{r,s}}} dz_1 \cdots dz_r =$$

$$\underbrace{G_r(z; 2/\alpha_+^2) = \prod_{i \neq j} \left(1 - \frac{z_i}{z_j}\right)^{\alpha_+^2/2}}_{\prod_{i \neq j} (z_i - z_j)^{\alpha_+^2/2} \prod_{i=1}^r z_i^{(1-r)\alpha_+^2/2}} \Big\} \text{ multivalued}$$

$$\times \underbrace{\prod_{i=1}^r z_i^s}_{\text{Jack poly}} \prod_{m \geq 1} \exp \left(\alpha_+ \frac{a_{-m}}{m} \underbrace{\sum_i z_i^m}_{\text{power sum}} \right) \exp \left(-\alpha_+ \frac{a_m}{m} \sum_i z_i^{-m} \right) \frac{dz_1 \cdots dz_r}{z_1 \cdots z_r} \Big|_{F_{\alpha_{-r,s}}}$$

Multivalued function $G_r(z; 2/\alpha_+^2)$ is well known integration kernel. Defines inner product with respect to which Jack symmetric polynomials are orthogonal.

Symmetric polynomials

- Ring of symmetric polynomials can be considered in either finite or infinite number of variables:
 - ▶ $\Lambda_r = \mathbb{C}[z_1, \dots, z_r]^{\mathfrak{S}_r}$
 - ▶ $\Lambda = \varprojlim_r \Lambda_r$
- Set variables to 0 to project down to smaller rings:

$$\begin{aligned}\pi_{r',r} : \Lambda_{r'} &\rightarrow \Lambda_r \\ f(z_1, \dots, z_{r'}) &\mapsto f(z_1, \dots, z_r, 0, \dots, 0),\end{aligned}$$

where $r' > r$.

Symmetric polynomials: examples

Rings Λ_r, Λ admit many bases and generators:

- The power sums (seen in expansion of screening fields)
 $p_m = \sum_i z_i^m$ generate ring of symmetric polynomials:

$$\Lambda_r = \mathbb{C}[p_1, \dots, p_r],$$

$$\Lambda = \mathbb{C}[p_1, p_2, \dots].$$

- The elementary symmetric polynomials

$$e_m = \sum_{1 \leq i_1 < \dots < i_m} z_{i_1} \cdots z_{i_m}$$

are also generators:

$$\Lambda_r = \mathbb{C}[e_1, \dots, e_r],$$

$$\Lambda = \mathbb{C}[e_1, e_2, \dots].$$

Symmetric polynomials: examples

The monomial symmetric polynomials form basis of Λ_r and Λ :

$$m_\lambda = \sum_{\sigma} z_1^{\lambda_{\sigma(1)}} z_2^{\lambda_{\sigma(2)}} \cdots ,$$

where σ runs over all distinct permutations of $\lambda = [\lambda_1, \lambda_2, \dots]$.

Examples:

- $m_{[2,1]}(z_1, z_2) = z_1^2 z_2 + z_1 z_2^2$
- $m_{[2,1]}(z_1, z_2, z_3) = z_1^2 z_2 + z_1 z_2^2 + z_1^2 z_3 + z_1 z_3^2 + z_2^2 z_3 + z_2 z_3^2$
- $m_{[s^r]}(z_1, \dots, z_r) = z_1^s z_2^s \cdots z_r^s$
- if $\ell(\lambda) > r$, then $m_\lambda = 0$ in Λ_r

Jack symmetric polynomials

Jack symmetric polynomials P_{λ}^{κ} are characterised by two properties:

- Upper triangular in symmetric monomials with respect to the dominance ordering of partitions:

$$P_{\lambda}^{\kappa}(z) = \sum_{\mu \leq \lambda} u_{\mu, \lambda}^{\kappa} m_{\mu}(z), \quad u_{\lambda, \lambda}^{\kappa} = 1,$$

where $\mu \leq \lambda$ if $|\mu| = |\lambda|$ and

$$\sum_{i=1}^n \mu_i \leq \sum_{i=1}^n \lambda_i, \quad \forall n \geq 1.$$

- Jack polynomials mutually orthogonal with respect to the inner product

$$\langle f, g \rangle_r^{\kappa} = \int_{\Gamma} G_r(z; \kappa) f(z_1^{-1}, \dots, z_r^{-1}) g(z_1, \dots, z_r) \frac{dz_1 \cdots dz_r}{z_1 \cdots z_r}.$$

Jack symmetric polynomials

Important properties of Jack polynomials:

- “Norm square” $\langle P_{\lambda}^{\kappa}, P_{\lambda}^{\kappa} \rangle_r^{\kappa}$ given by known combinatorial formulae.
- Jack polynomials satisfy many powerful identities.

Simple examples of Jack polynomials:

- Elementary symmetric polynomials are Jack polynomials

$$e_k = m_{[1^k]} = P_{[1^k]}^{\kappa}, \forall \kappa.$$

- In Λ_r

$$P_{[s^r]}^{\kappa}(z_1, \dots, z_r) = m_{[s^r]}(z_1, \dots, z_r) = \prod_{i=1}^r z_i^s, \quad \forall \kappa.$$

Explicit singular vector formulae

Thm: [Mimachi, Yamada], [Tsuchiya, SW]

Explicit formulae for all singular vectors of Fock spaces:

$$S_+^{[r]}|\alpha_{r,-s}\rangle = \text{const} \cdot \rho \left(\mathbf{P}_{[s^r]}^{2/\alpha_+^2} \right) |\alpha_{-r,-s}\rangle, r \geq 1, s \geq 0,$$

where $\rho(\mathbf{p}_m) = a_{-m}2/\alpha_+$.

Equivalent formulae exist for S_- .

Proof: Use the inner product of Jack polynomials to evaluate action of screening operators.

Singular vector of Heisenberg vertex operator algebra

$$|\chi\rangle = S_+^{[p_+-1]}|\alpha_{p_+-1,1-p_-}\rangle.$$

Finding minimal model conformal weights

- Action of the singular field $\chi(z)$ trivial on minimal model reps.
 - ▶ Find the zeros in τ of the matrix element

$$\langle \tau | \chi(z) | \tau \rangle,$$

where $\tau \in \mathbb{C}$, $|\tau\rangle$ generating vector of F_τ and $\langle \tau | a_{-n} = 0, n \geq 1$ dual vector of $|\tau\rangle$.

- Matrix element evaluable by known identities of Jack polynomials

$$\langle \tau | \chi(z) | \tau \rangle = \text{const} \prod_{(r,s) \in \text{Kac}} (h_\tau - h_{r,s}),$$

where $h_\tau = \frac{\tau}{2}(\tau - \alpha)$ conformal weight of $|\tau\rangle$ and

$$h_{r,s} = \frac{(p_+ s - p_- r)^2 - (p_+ - p_-)^2}{4p_+ p_-}.$$

Summary and outlook

- Today's result: Symmetric polynomials can be used to “quite easily” classify the irreducible minimal model representations.
- Essentially the same method also works for free field realisations of affine $\mathfrak{sl}_2$ [Ridout, SW] and should work in many more cases.
- This method was a crucial tool in the classification of the irreducible representations of certain logarithmic extensions of the minimal models [Tsuchiya, SW].

Punch line

Free field theories have deep connections to symmetric polynomials and symmetric polynomials have rich well known combinatorics $\Rightarrow$ Free field realisations are even more useful than previously assumed!

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Thank you for coming!